Higher National Diploma in Quantity Surveying
Second Year, 1st semester Examination - 2019
QS 21023–Building Economics 1

Instructions for Candidates:

Part A is Compulsory	No. of questions : 06
Answer 03 Questions from Part –B	No. of pages : 05
Answer for each question must be started in new page	**Time: Three (3) hours**

PART A

1. Royal Institute of British Architectures (RIBA) plan of work is introduced for project formulation.

   1.1. Name the main stages of RIBA plan of work. (04 marks)

   1.2. Conceptual design stage is basically considered concept and methodology for development of design. Explain briefly conceptual design stage. (04 Marks)

   1.3. Appraisal (Inception) stage is involved more professionals and expertise to evaluate project output. Describe the role of professionals and expertise in appraisal stage. (04 marks)

   1.4. Quantity Surveyor (QS) plays a major role in tender action stage of RIBA plan of work. Explain QS's role in this stage. (04 marks)

   1.5. What are the benefits for the stakeholders of a project by application of RIBA plan of work for formulation of the project? (04 marks)

2. Construction industry in Sri Lanka is monitored by Construction Industry Development Authority (CIDA) under the Ministry of Housing and Construction.

2.1. Who are the direct and indirect stakeholders in construction industry? (04 marks)

2.2. Name the four regulator bodies and their roles in construction industry. (04 marks)

2.3. Explain briefly main categories of civil engineering contractors graded by CIDA for the registration. (04 marks)

2.4. If a Contractor required registering his company for Civil Engineering Contracts in CIDA, what are the qualifications the Contractor should provide to CIDA? (04 marks)

2.5. Construction industry faces difficulties to recruit skill workforce. Define skill workforce and their contribution to construction industry. (04 marks)

PART B

3. Positive economic growth of a country indicated the upliftment of living condition with relative to the previous year. Gross Domestic Product (GDP) and Gross National Product (GNP) and Per Capita Income are the other indicators for the measuring of economic development.

3.1. What are the three approaches in calculation of GDP of a country? (03 marks)

3.2. Calculate the GDP in year 2017 by Expenditure approach. Following data are provided.

Indirect Business Tax receivable	Rs. Billion	50
Income receipt from the rest of world	Rs. Billion	85
Gross Private Domestic Investment	Rs. Billion	312
Personal income Interest	Rs. Billion	270
Imports	Rs. Billion	113
Export	Rs. Billion	99
Wages as income by rest of the world	Rs. Billion	360
Personal Consumption expenditure	Rs. Billion	560
Income payment by the rest of the world	Rs. Billion	72
Government Expenditure	Rs. Billion	235

GDP = C + I + G + (X-M)

C	Consumer Expending	
I	Gross Private Domestic Investment	
G	Government Expenditure	
(X – M)	Net Export	(06 marks)

3.3. Explain briefly how the construction industry contributes the national economy. (03 marks)

3.4. Price floors have been widely legislated for agricultural products such as rice, milk...etc., as a way to support the income of farmers. Explain the meaning of price floor with supply and demand curve. (04 marks)

3.5. Many countries maintain a minimum wage and age limit for the workers. What is meant by minimum wages and age limit? (04 marks)

4. Estimating process is involved with various factors such as standard and specification, historical data, expert judgment, economic condition, overheads and profit margin…etc.

4.1. Explain the preliminary or approximate estimates and mention the purpose of preparing this estimate. (03 marks)

4.2. What are the qualifications of an estimator? (03 marks)

4.3. Historical data is given below for preparation of approximate estimate for the proposed domestic single-story house. Calculate the approximate estimate for 750 m^2 domestic single-story house, proposed to construct in year 2020.

Assume	Architectural and design cost	- 3% of Building cost
	Utility cost	- 4% of Building cost
	Landscaping and others facilities	- 6% of Building cost
	Supervision cost	- 5% of Overall cost
	Contract administration and management	- 3% of overall cost

Year	Building Cost per m² in Rs
2010	33,000
2012	38,000
2014	41,500
2016	48,000

(08 marks)

4.4. Accuracy of detail estimate is very much important for planning of financial proposal for a project. What are the factors affected to the accuracy of detail estimates? (04 marks)

4.5. Explain the difference between Engineer's Estimate and Bid Estimates. (02 marks)

5. Poor attention on Cost, Quality and Time is caused cost overrun of the project. Proper planning of resources, time frame of project and quality standard and specification including material and workmanship is essential to reduce cost overrun.

5.1. Explain briefly cost overrun of a construction project. (04 Marks)

5.2. What are the basic steps of Quality Controlling (QC) of a construction project? (04 marks)

5.3. Explain the methods of monitoring financial and physical progress of a project. (04 marks)

5.4. Site Planning is a key activity before commencement of a construction. What are the important considerations for site planning? (04 marks)

5.5. Explain briefly occupational health and safety precaution is applied in a construction project. (04 marks)

6. Foreign investment is essential for the economic development of Sri Lanka. Government implemented a Board of Investment (BOI) to promote the foreign investment.

6.1. Explain the responsibility of BOI to promote the foreign investment. (04 marks)

6.2. What do you mean by the green building concept? List the factors which affect for making a green building. (04 marks)

6.3. Explain difference between the exports and re-export. (04 marks)

6.4. Reuse, Recycle, Reduce (3R) concept is help to minimize environmental pollution. Explain this statement. (04 marks)

6.5. What is meant by EIA (Environmental Impact Assessment) and IEE (Initial Environmental Evaluation) IEE? (04 marks)